



#### FUTURE FLIER

*The Aerion AS2 is one of a number of supersonic passenger aircraft on the drawing board in the second decade of the 21st century. Backed by Airbus, it is a business jet intended to fly at up to Mach 1.5, carrying 12 passengers. The AS2 is currently projected to achieve its first flight in 2019 and enter service in 2023.*



and contemporaries in its stealth features and its ability to cruise at supersonic speed, without use of afterburners – all other fighters could only “go supersonic” in short bursts because their afterburners used up fuel so quickly.

But the F-22 was also very expensive indeed. With only 187 built before production halted in 2011, the cost was estimated at over \$400 million per aircraft. Any discussion regarding the possible value for money for the F-22 was controversial as with the ending of the Cold War, it was difficult to see when such an advanced fighter aircraft would be needed.

Arguably, the Russians or the Chinese still had the capacity to develop aircraft at the cutting edge of this technology and the potential will to use it in a major war.

However, the cheaper, mass-produced F-35 – achieving combat readiness in 2016 after a slow and troubled

development – seemed fully adequate to cope with the air superiority task as well as a wide range of other fighter roles. Despite huge cost overruns, the unit price of an F-35 was estimated at \$106 million, less than one third of the cost of an F-22, but there are huge differences in capability between the two aircraft. The compromise between cost and technology looked more likely to prove successful in the long run. The lesson to be learned was that developments in aviation could not be safely predicted on the basis of what was technologically feasible. At the time of maximum faith in technological progress – an era that effectively ended in the 1970s – it was assumed that anything that could be done, would be done. By the 21st century, a more utilitarian logic was in the ascendant. If a development in aviation could perform a required function for a reasonable price, it would happen. Whether funded by governments or by business, new aircraft projects that did not meet these criteria were destined to run into dead ends.

Space exploration might be the only area where the technological imperative could still predominate. It remained the open frontier where, at least in theory, boundless possibilities existed for new achievement and funding might materialize for almost any project that sufficiently caught the public imagination.

#### CREW DRAGON

*The SpaceX Dragon 2, or Crew Dragon, is a manned space vehicle developed from the unmanned Dragon 1, which has been ferrying cargo to the International Space Station (ISS) since 2012. The Crew Dragon here stands on its launch pad.*